KIRAN S M

Email:kiranmanikkath20@gmail.com **LinkedIn:** Kiran S M
Mobile: 7736804506 **Portfolio:**

Profile Summary

Electronics and Communication Engineering student with a keen interest in coding, web development, and embedded systems. Currently exploring projects involving Raspberry Pi and Arduino, with a strong drive to develop innovative, technology-driven solutions. Passionate about integrating hardware and software for real-world applications, continuously enhancing technical skills. Interested in contributing to challenging projects and gaining hands-on experience in the field.

Academic Qualification

Course	Year	Institute/Board	CGPA/%
Bachelor of Technology in Electronics and Communication	2026	NSS College of Engineering, Palakkad, Kerala	8.2 (Till 5th Sem)
12^{th}	2022	HSE	97.25%
10^{th}	2020	CBSE	93.2%

Internship

Role : Intern
Organisation Name : Acutro Technologies Pvt. Ltd, Thiruvananthapuram, Kerala, **20 June 2024 - 28 June 2024**
Description : Familiarised with the concepts of IoT and Robotics. Developed a python-based voice chatbot using OpenCV. Gained experience in microcontroller programming using embedded C.

Projects

1. Title : **Smart Sleep Pad**, Mar 2025 **(Mini Project)**
Description : Designed and implemented a smart pad to monitor sleep quality using FSRs, MAX30102 and MPU6050, with real-time data transfer to a web app via ESP32 and MQTT. Integrated features include heart rate monitoring, body movement tracking, sleep cycle analysis, vibration-based alarms, and Firebase storage. Tools used: ESP32, MAX30102, MPU6050, FSR Sensors, Vibration Motors, Firebase, MQTT.

2. Title : **Line Following Robot**, July 2024 **(Course Project)**
Description : Designed and implemented an autonomous robot using Arduino UNO that follows black lines on a surface through IR sensors and motor drivers. Tools used: Arduino UNO, IR Sensors, Motor Driver, DC Motors.

3. Title : **Up Counter Circuit**, Dec 2023 **(Course Project)**
Description : Built an electronic up counter circuit using digital logic ICs to count events from 0 to 9 and display them on a seven-segment display, with reset functionality. Tools used: Digital Logic ICs, Seven-Segment Display.

4. Title : **Automatic Light Fence Circuit with Alarm**, Nov 2024 **(Course Project)**
Description : Developed a light-based fence security system with an alarm using an operational amplifier and LDR sensors for intrusion detection and response. Tools used: Operational Amplifier, LDR Sensor, Buzzer.

5. Title : **Supreme Leader Website**, Nov 2024 **(Hackathon Project)**

6. Title : **Wax Pen**, Dec 2023 **(Design Project)**

Technical Skills

- Programming Languages: Python,C/C++,8051 Assembly language
- Frontend: React,HTML,CSS,Javascript
- Simulation and Design: Proteus,LTspice,Verilog
- Libraries: Matplotlib,Numpy

Certifications

- Introduction to Programming in C, **IIT Kanpur,Jan-Mar 2023**
- The Joy of Computing using Python, **IIT Madras,Jan-Apr 2024**
- Career Essentials in Generative AI, **Microsoft & LinkedIn,Aug 2024**
- Communication Skills, **TCS iON,June 2025**

Workshops, Competitions and Events

- Ethical Hacking,**NIT Calicut,March 2023**
- Digital Design,**IETE NSSCE,Dec 2023**
- Participated in Microcontroller-Based Embedded System Design Hands-on event,**IEEE,Oct 2024**
- Participated in Startup Bootcamp of Kerala Startup Mission,**APJAKTU,July 2024**

Achievements

- **3rd position** in the ideathon jointly organized by KSCSTE, IIT Palakkad and NSS College of Engineering
- Selected among the top 10 teams nationally for the Development Phase of the IEEE MYOSA 2.0 Innovation Challenge by IEEE Sensors Council.

Languages

• English • Hindi • Malayalam • Tamil

Hobbies and Extra-Curricular Activities

• Cinema • Cricket • Dance • Music